

SPY: SPDR S&P 500 ETF, which is designed to mimic the S&P 500;

SPYV: SPDR S&P 500 Value ETF, which tracks the S&P 500 value index; and

SPYG: SPDR S&P 500 Growth ETF, which replicates the S&P 500 growth index.

These low-cost index ETFs are tradeable, unlike the Fama–French portfolios. They belong to the SPDR (pronounced “spider”) family of ETFs sponsored by State Street Global Advisors.

Our benchmark factor regression for fund i (but I avoid the i subscripts to make the notation clearer) is

$$r_{t+1} = \alpha_t + \beta_{SPY,t} SPY_{t+1} + \beta_{SPYV,t} SPYV_{t+1} + \beta_{SPYG,t} SPYG_{t+1} + \varepsilon_{t+1}, \quad (10.14)$$

where we impose the restriction

$$\beta_{SPY,t} + \beta_{SPYV,t} + \beta_{SPYG,t} = 1,$$

so that the factor loadings, or factor weights, sum to one. The factor weights on the right-hand side of equation (10.14) constitute a replicating portfolio for fund i .

The main idea with style analysis is that we use actual tradeable funds in the factor benchmark. I used SPDR ETFs in equation (10.14), but I could have used other ETFs or index mutual funds for the benchmark portfolio.

Note the timing in equation (10.14). The weights are estimated using information up to time t . The return of the fund over the next period, $t+1$, is equal to the replicating portfolio formed at the beginning of the period at time t plus a fund-specific residual, ε_{t+1} , and the fund alpha, α_t for that period. The weights can change over time. Equation (10.14) asks, “Can we find a robot that makes time-varying investments in SPY, SPYV, and SPYG that, together, match the returns of Buffett?”

Figure 10.5 graphs the factor weights (or style weights) of the four funds. I estimate the factors using data over the previous sixty months, $t - 60$ to t , to form the benchmark weights at time t . In addition to imposing that the factor weights sum to one, I also constrain the factor weights to be all positive (so there is no shorting). The first factor weight is estimated at January 2006.

Panel A of Figure 10.5 shows that LSV is merely a combination of the market (SPY) and value (SPYV). Fidelity Magellan starts off in 2006 as a combination of all three ETFs but, at the end of 2012, ends up being all growth (SPYG). Goldman’s growth fund is mostly market exposure (SPY) and growth (SPYG) at the beginning of the sample and at the end of the sample is just growth (SPYG). Buffett’s factor exposure is the most interesting. He starts off in 2006 being strongly value (SPYV). During the financial crisis, he switches styles to become growth. Then as the crisis subsides, he goes back to being a strong value manager.